DWDM - CSA1603 Cos- ATL-02

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Q1. APRIORI ALGORITHM

Pseudo code:

START
Input transcation dataset
set minimum support
Find all individual items
Count support of each item
Remove items below minimum support
Keep the remaining items as frequent itemsets
WHILE Frequent itemsets are available
Generate candidate itemsets
Count support of each candidate
Remove candidates below minimum support
Keep the remaining candidates as frequent itemsets
END WHILE
Display all frequent itemsets
STOP.

Candidate Generation

From the dataset:

1. itemsets:

{mobile}

{earphones}

{charger}

Generate:

2. itemsets:

{mobile, earphones}

{mobile, charger}

{earphones, charger}

If the support is sufficient, generate:

3. itemsets:

{mobile, earphones, charger}

(2) FP-Growth Algorithm

START

Input borrowing transctions

Set minimum support

Count the frequency of each book

Remove books below minimum support

Sort remaining books by frequency

Create the FP-tree

Create the Header Table

For each book in the Header Table

Find its conditional pattern base

Create a conditional FP-tree

Generate frequent itemsets

END for

Display all frequent itemsets

STOP

FP-Tree construction

First count the books.

Python → 4

machine Learning → 4

Data science → 3

Then arrange the books accoriding to their frequency and insert each Transcation into the FP-Tree

For example:

Root
├→ Python
│   └→ Machine Learning
└→ Data science
     └→ machine Learning

③ Association Rule Generation

Pseudo code

```
START
  Input frequent itemsets
  set minimum confidence
ROR each frequent itemset
      Generate possible rules
    for each possible Rules
      Calculate confidence
  IF  confidence >= minimum confidence
        Keep the rule
    ELSE
        Remove the rule
      END IF
```

END FOR

END FOR

Display valid association rules

STOP

Rule Generation

From

{Internet, streaming}

Possible rules

Internet → streaming

Streaming → Internet

FROM : {streaming, cloud storage}

Possible rules:

{streaming → cloud storage

cloud storage → streaming

Confidence:

Confidence tells us how often the second service is purchased when the first service is purchased.

Example:

Confidence (Internet → streaming)

= support (Internet, streaming) / support (Internet)

= 3/4

⇒ 75%

If the minimum confidence is 70%. Thus this rule is accepted.

(4) Constraint - Based Frequent Itemset mining

Dataset : student course Enrollment

Pseudocode:

START

Input student course transactions

set constraint = "python"

Find all individual courses

Generate candidate itemsets

For each candidate itemsets

Generate candidate itemsets

For IF itemset contains "python"

Calculate support

IF support >= minimum support

Keep the itmset

ELSE Remove the itemset

END IF

ELSE Remove the itemset

END FOR

Generate larger itemsets containing "python"

Repeat until no more itemsets can be generated

Display frequent itemsets containing "python"

STOP

Candidate Generation

From the dataset:

{python}

{python, Databases

{python, AI}

{python, Database, AI}

only itemsets containing python are considerded.

⑤ multilevel Association Rule mining

Dataset: Hospital Treatment Hierachy

Pseudocode:

```
START
  Input treatment transactions
  Input treatment hierachy
 Set minimum support and confidence
Find Frequent treatments at the top level
 Generate Frequent itemsets
  For each hierarchy level
       Generate Frequent itemsets at that level
    Calculate support
  Remove itemsets below minimum support
    Generate Association rules
      calculate confidence
 Keep rules above minimum confidence
    END For
 STOP
```

Treatment Hierarchy

```
Healthcare
 ├─ cardiology
 │     └> ECG
 └─ orthopedics
```